



N-Gram Model

Example Problem

Consider the following small corpus:

- AI changes world
- AI improves healthcare
- AI changes healthcare

Find the probability for AI improves healthcare using N-Gram Model (Bi-gram Model)

Step 1: Add Sentence Boundaries

- <s> AI changes world </s>
- <s> AI improves healthcare </s>
- <s> AI changes healthcare </s>

Step 2: Unigram Counts

Word	Count
<s>	3
AI	3
changes	2
world	1
Improves	1
Healthcare	2
</s>	3
Total Tokens	15

Step 3: Bigram Counts

Word	Count
<s> AI	3
AI changes	2
AI Improves	1
Changes world	1
world </s>	1
Changes Healthcare	1
Improves Healthcare	1
Healthcare </s>	2

Step 4: Bigram Probability Formula

$$P(w_n|w_{n-1}) = \frac{Count(w_{n-1}, w_n)}{Count(w_{n-1})}$$

Step 5: Compute Probabilities

Probabilities	Value
$P(\text{AI} \mid \langle s \rangle)$	$3/3$
$P(\text{changes} \mid \text{AI})$	$2/3$
$P(\text{improves} \mid \text{AI})$	$1/3$
$P(\text{World} \mid \text{changes})$	$1/2$
$P(\text{healthcare} \mid \text{changes})$	$1/2$
$P(\text{healthcare} \mid \text{improves})$	$1/1$
$P(\langle /s \rangle \mid \text{World})$	$1/1$
$P(\langle /s \rangle \mid \text{healthcare})$	$2/2$

Step 6: Sentence Probability

Find the probability of: **AI changes healthcare**

Using Bigram Model:

$$P(\text{AI changes healthcare})$$

$$= P(\text{AI} | < s >) \times P(\text{changes} | \text{AI}) \times P(\text{healthcare} | \text{changes}) \\ \times P(</s> | \text{healthcare})$$

- $P(\text{AI} | < s >) = \frac{3}{3} = 1, P(\text{changes} | \text{AI}) = \frac{2}{3}$
- $P(\text{healthcare} | \text{changes}) = \frac{1}{2}, P(</s> | \text{healthcare}) = 1$



Final Calculation $= 1 \times \frac{2}{3} \times \frac{1}{2} \times 1 = \frac{1}{3} = 0.33$

$P(\text{AI changes healthcare}) = 0.33$

$P(\text{AI improves healthcare}) = ?$

Example: 2

Given Corpus

- Space exploration inspires innovation
- Space research inspires scientists
- Space exploration advances science

Find the probability for “Space exploration advances science”
using N-Gram Model (Tri-gram Model)

Trigram Probability Formula

$$P(w_n | w_{n-2}, w_{n-1}) = \frac{\text{Count}(w_{n-2}, w_{n-1}, w_n)}{\text{Count}(w_{n-2}, w_{n-1})}$$